



EMERITUS

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Masterclass 2: Building Your GitHub Profile & Publishing Projects



What This Masterclass is About?



Understand the fundamentals of GitHub and version control



Upload projects and manage repositories

Learning Objectives

Create, organize, and maintain a professional GitHub profile

Upload projects and manage repositories

Document projects effectively using README files and commit messages

Publish and share data analysis activity from Masterclass 1

Git

Version Control System is a tools that helps to track changes in code

Git is a Version Control System. It is :

popular

free & Open Source

fast & scalable

Github

Website that allows developers to store and manage their code using Git.

<https://github.com>

Why is GitHub useful?

Version control

- No more.... *script*, *script_v2*, *script_v2_10-18* edits, etc.
- Can always go back to any previous version!

Collaboration

- Integrates edits from multiple people
- Keeps record of who did what and when

Documentation

- All scripts in a single place with Readme files

Reproducibility

- You and others can recreate and/or modify an script you created, even many years later

Visibility

- Showcase your work to potential employers / schools
- Promote your lab's achievements and facilitate the use of your pipelines

Setting Up GitHub

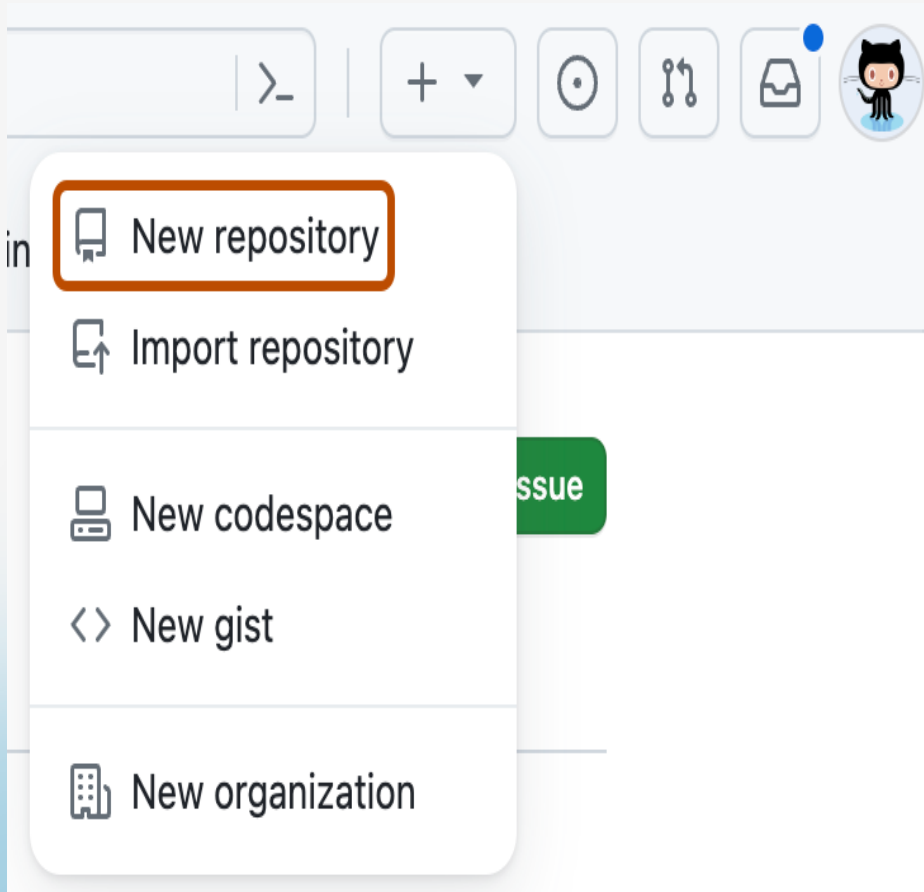


Create an account at [GitHub.com](https://github.com)

Customize profile: Add a bio, profile picture, and pinned repositories

Explore GitHub features: Repositories, commits, branches, and pull requests

Creating a Repository



Click on "New Repository"

Name the repository and add a description

Initialize with a README

Upload project files (data, scripts, visuals)

Documenting the Project

Project Documentation



README.md essentials:

Project title & description

Installation & usage instructions

Dataset explanation (if applicable)

Author information & acknowledgments

Commit messages: Best practices for clarity and tracking changes

Setting up Git

Visual Studio Code

Windows (Git Bash)

Mac (Terminal)

git --version

Configuring Git

```
git config --global user.name "My Name"
```

```
git config --global user.email "someone@email.com"
```

```
git config --list
```

Clone & Status

Clone - Cloning a repository on our local machine

git clone <- some link ->

status - displays the state of the code

git status

Types of Status

untracked

new files that git doesn't yet track

modified

changed

staged

file is ready to be committed

unmodified

unchanged

Add & Commit

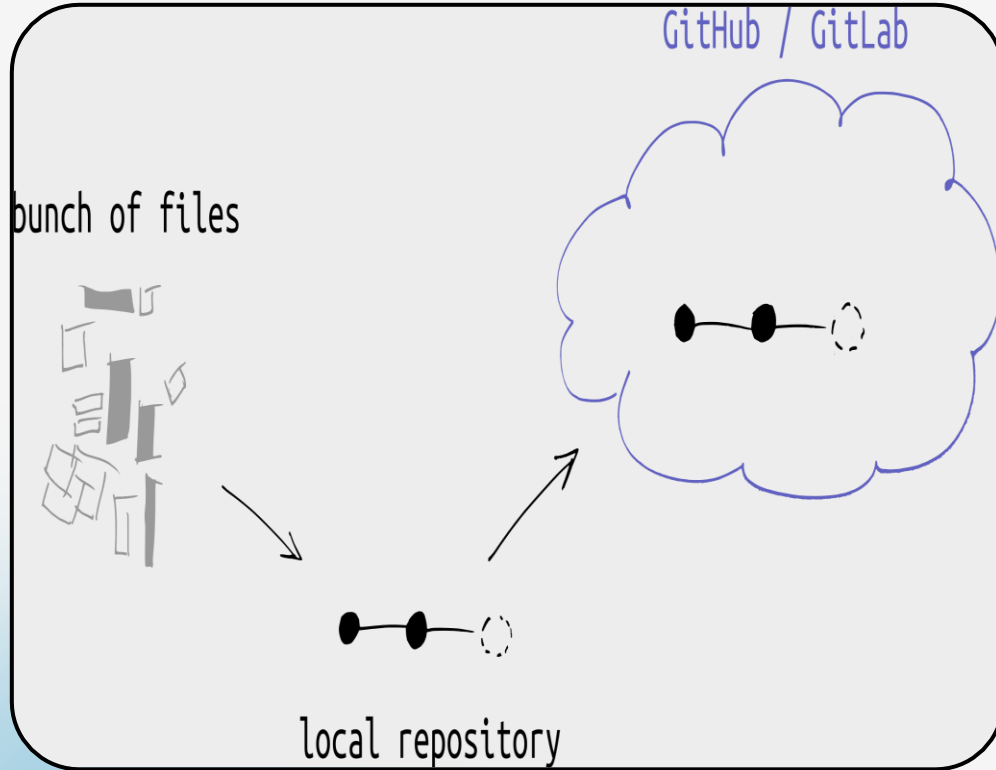
add - adds new or changed files in your working directory to the Git staging area.

git add <- file name ->

commit - it is the record of change

git commit -m "some message"

Publishing and Sharing



Push updates to GitHub using Git or the web interface

Share repository link on portfolios, LinkedIn, and professional networks

Push Command

push - upload local repo content to remote repo

git push origin main

Init Command

init - used to create a new git repo

git init

git remote add origin <- link ->

git remote -v (to verify remote)

git branch (to check branch)

git branch -M main (to rename branch)

git push origin main

Edit, stage/add, commit, push, repeat!

create / edit files

git push

- Sends your changes to the remote repository (on GitHub)

git add -A

- Stage files you've added or edited
- Can also use 'add' command to stage specific files

git commit -m <commit notes>

- Commits the changes to your local repository
- Lets you document the latest changes you've made

```
PD_schNaseq --bash --80x26
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

  git config --global --edit

After doing this, you may fix the identity used for this commit with:

  git commit --amend --reset-author

3 files changed, 1025 insertions(+)
create mode 100644 .gitignore
create mode 100644 data_detekt.txt
create mode 100644 run_secret.Rmd
matthew-wp-2:PD_schNaseq rajlali$ git push
Counting objects: 3, done.
Delta compression using up to 8 threads.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (3/3), 512.56 KiB | 8.54 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
To https://github.com/rajlali/PD_schNaseq.git
   95f6ed..3099d81 master -> master
matthew-wp-2:PD_schNaseq rajlali$
```

```
PD_schNaseq --bash --80x24
Last login: Tue May 27 16:50:08 on ttys002
matthew-wp-2 ~ - rajlali$ cd Desktop/PD_schNaseq
matthew-wp-2:PD_schNaseq rajlali$ git add -A
matthew-wp-2:PD_schNaseq rajlali$
```

```
PD_schNaseq --bash --80x24
Last login: Tue May 27 16:46:30 on ttys002
matthew-wp-2 ~ - rajlali$ cd Desktop/PD_schNaseq
matthew-wp-2:PD_schNaseq rajlali$ git add -A
matthew-wp-2:PD_schNaseq rajlali$ git commit -m "added initial metadata and analysis script"
[master 95f6ed] added initial metadata and analysis script
1 file changed, 1025 insertions(+)
create mode 100644 .gitignore
create mode 100644 data_detekt.txt
create mode 100644 run_secret.Rmd
matthew-wp-2:PD_schNaseq rajlali$
```

Expected Outcomes



A structured, professional GitHub profile

A published data analysis project from Masterclass 1

Confidence in using GitHub for version control and collaboration

Enhanced personal branding in the digital world

Conclusion



Recap of key learnings

Q&A and hands-on support